

8. Even More Integral Applications

8.3. Moments and Center of Mass

8.3 Learning Objectives

- Understand systems of individual masses and laminae
- Find the moment of a system about a given axis
- Find the center of mass of a system of n masses
- Find the centroid of a system with one continuous mass

8.3.1. Motivation

Suppose we have 2 masses (of different mass) supported on a massless rod (picture 2 heavy weights held up by a thin wire in comparison. The wire has essentially no weight compared to the weights). Where could we put a **fulcrum** so that the system is balanced?

In Figure 1 Alexander Calder brings this concept to life through his art. He has created many pieces involving the balancing of weights held with wire as mobile sculptures, this one is called “Black and Yellow Dots in the Air”. He was originally trained as a mechanical engineer, and had some sculptors in the family.

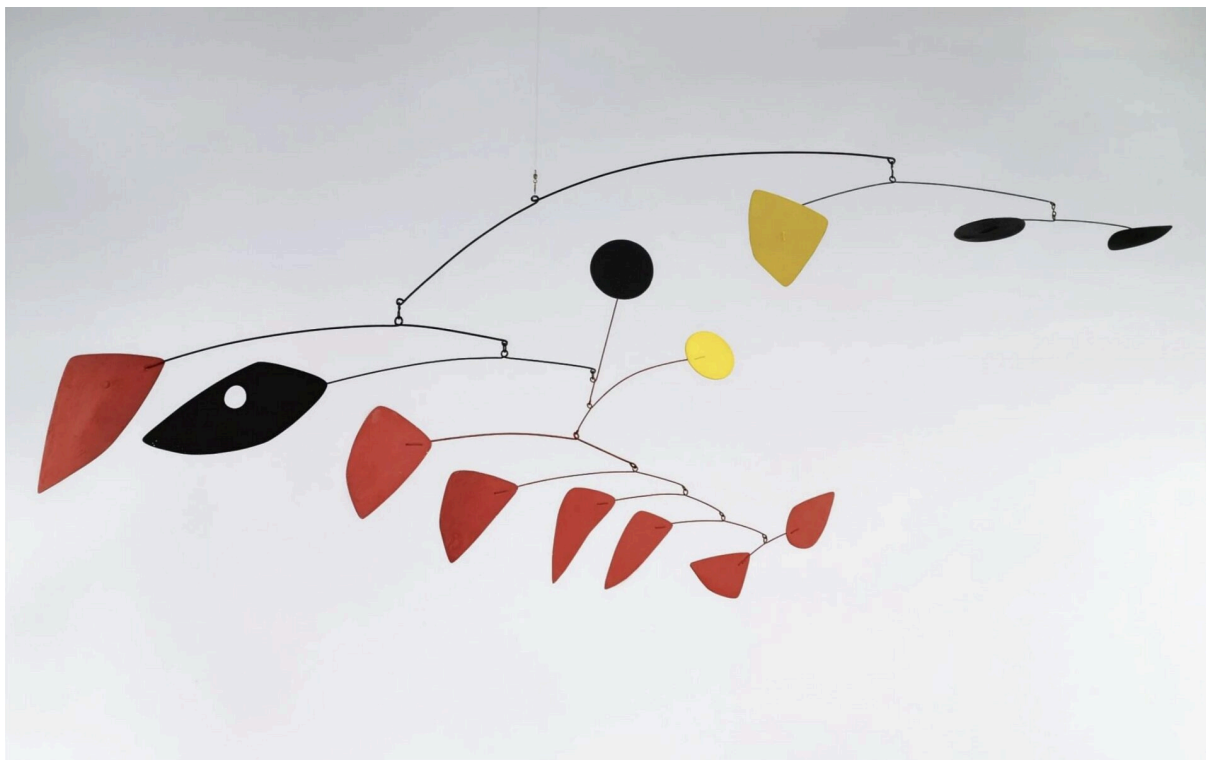
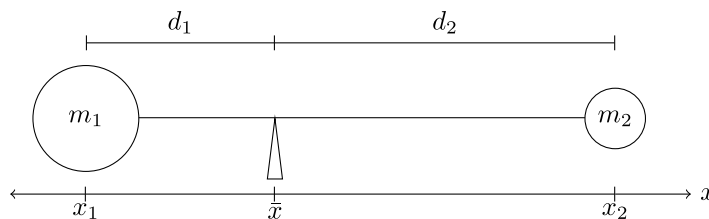


Figure 1: Black and Yellow Dots in the Air by Alexander Calder

In the physics sense, we need to balance the torque about a specific point.



In the 1 dimensional case, we need to balance the mass times the distance on both sides of our fulcrum. That is, if

$$m_1 d_1 = m_2 d_2$$

the system will be balanced. From the sketch we see that $d_1 = (\bar{x} - x_1)$ and $d_2 = (x_2 - \bar{x})$, and doing a bit of algebra we can solve for $\bar{x}$, the location to put the fulcrum.

$$\begin{aligned} m_1 d_1 &= m_2 d_2 \\ \Rightarrow m_1(\bar{x} - x_1) &= m_2(x_2 - \bar{x}) \\ \Rightarrow m_1 \bar{x} - m_1 x_1 &= m_2 x_2 - m_2 \bar{x} \\ \Rightarrow m_1 \bar{x} + m_2 \bar{x} &= m_2 x_2 + m_1 x_1 \\ \Rightarrow \bar{x}(m_1 + m_2) &= m_2 x_2 + m_1 x_1 \\ \Rightarrow \bar{x} &= \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}. \end{aligned}$$

8.3.2. 1-Dimensional Moment and Center of Mass for Point Masses

Definition 8.3.1 Moment and Center of Mass

We say that the **moment** of a system about the origin is given by

$$M = \sum_{k=1}^n m_k x_k,$$

for a system with n objects, where m_i represents each objects mass and x_i represents the *signed* distance from the origin.

We say that each $m_i x_i$ is a **moment**.

The location

$$\bar{x} = \frac{M}{\sum_{k=1}^n m_k} = \frac{\sum_{k=1}^n m_k x_k}{\sum_{k=1}^n m_k}$$

is called the system's **center of mass**. In other words, the moment divided by the total mass is the center of mass.

Example 8.3.1

Given point masses $m_1 = 8$, $m_2 = 4$, and $m_3 = 7$ located on the x -axis at $a_1 = 3$, $x_2 = -1$, and $x_3 = 6$, respectively, find the moment of the system about the y -axis, and the center of mass.

Solution

The moment about the y -axis is

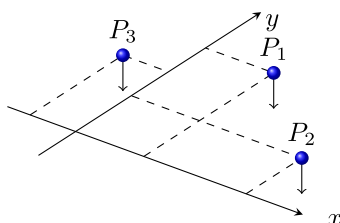
$$M = \sum_{k=1}^3 m_k x_k = 8(3) + 4(-1) + 7(6) = 62.$$

The total mass is $m = 8 + 4 + 7 = 19$. So the center of mass is at

$$\bar{x} = \frac{M}{m} = \frac{62}{19}.$$

8.3.3. 2-Dimensional Moment and Center of Mass for Point Masses

Now we move to looking at point masses on a plane (instead of on a line). Consider the moments about the x -axis and y -axis for objects in two dimensions (and also z -axis if in 3-D).



Let $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$, and $P_3 = (x_3, y_3)$.

Then, how far is P_1 away from the x -axis? It is y_1 away from the x -axis! (Justify to yourself why the distance is the y value and not the x value of the point)

How far is P_1 away from the y -axis? It is x_1 away from the y -axis!

We can then think about the moment in the x direction separate from the moment in the y direction!

Definition 8.3.2 Moments in 2-D

The **total moment of the system about the x -axis** is

$$M_x = \sum_{k=1}^n m_k y_k.$$

The **total moment of the system about the y -axis** is

$$M_y = \sum_{k=1}^n m_k x_k.$$

The **center of mass for the system** is $(\bar{x}, \bar{y})$ where

$$\bar{x} = \frac{M_y}{\sum_{k=1}^n m_k} = \frac{\sum_{k=1}^n m_k x_k}{\sum_{k=1}^n m_k}$$

$$\bar{y} = \frac{M_x}{\sum_{k=1}^n m_k} = \frac{\sum_{k=1}^n m_k y_k}{\sum_{k=1}^n m_k}$$

Example 8.3.2

Suppose $m_1 = 1$ is located at $(2, 3)$, $m_2 = 3$ is located at $(3, -4)$, and $m_3 = 4$ is located at $(-3, 1)$. Calculate the moments M_x and M_y and find the center of mass of the system.

Warning

M_x is the moment about the x -axis, meaning we are asking how far the points are from the x -axis and use the y values of the points for this! It gives the y value of the center of mass point!

The opposite is true for M_y .

Solution

First

$$M_x = \sum_{k=1}^3 m_k y_k = 1(3) + 3(-4) + 4(1) = -5$$

and

$$M_y = \sum_{k=1}^3 m_k x_k = 1(2) + 3(3) + 4(-3) = -1.$$

The total mass is $m = 1 + 3 + 4 = 8$. Then

$$\begin{aligned}\bar{x} &= \frac{M_y}{m} = -\frac{1}{8} \\ \bar{y} &= \frac{M_x}{m} = -\frac{5}{8}\end{aligned}$$

So, the center of mass is at $(-\frac{1}{8}, -\frac{5}{8})$.

8.3.4. Center of Mass of a Lamina

Definition 8.3.3

A solid flat object, called a **lamina**, of uniform density ρ which occupies a region $\mathcal{R}$ of the plane, has a center of mass called the **centroid** of $\mathcal{R}$.

How do we find the centroid of a lamina?

Suppose we have a simple region $\mathcal{R}$ defined under the curve $f(x)$ between a and b . Then using our usual method of taking finite sums to approximate what we want and taking the limit we cut the interval up into n sub-intervals with width Δx and height $f(x_i)$ where x_i is the value between each sub-interval. Then the mass of each strip is

$$\begin{aligned}\text{mass} &= \text{density} \cdot \text{area} \\ m_i &= \rho A(x_i) \\ m_i &= \rho f(x_i) \Delta x\end{aligned}$$

The moment about the x -axis (and remembering to think about how far away from the x -axis things are using y values, and how the strip is kind of like a point mass at the half of the height) is

$$\begin{aligned}
 M_x &\approx \sum_{i=1}^n m_i y_i = \sum_{i=1}^n (\rho f(x_i) \Delta x) \left(\frac{1}{2} f(x_i) \right) \\
 &= \frac{\rho}{2} \sum_{i=1}^n (f(x_i))^2 \Delta x
 \end{aligned}$$

And taking the limit as n goes to infinity we have

$$M_x = \int_a^b \frac{\rho}{2} (f(x))^2 dx$$

Next, the moment about the y -axis is

$$\begin{aligned}
 M_y &\approx \sum_{i=1}^n m_i x_i = \sum_{i=1}^n (\rho f(x_i) \Delta x) (x_i) \\
 &= \rho \sum_{i=1}^n (f(x_i) x_i \Delta x)
 \end{aligned}$$

And taking the limit as n goes to infinity we have

$$M_y = \int_a^b \rho f(x) x dx$$

Lastly, the total mass is what we get from adding up all the m_i and taking the limit as n goes to infinity so

$$m = \int_a^b \rho f(x) dx$$

For a lamina made up of a region $\mathcal{R}$ under the curve $f(x)$ on $[a, b]$ the centroid is the point

$$(\bar{x}, \bar{y}) = \left(\frac{\int_a^b \rho f(x) x dx}{\int_a^b \rho f(x) dx}, \frac{\int_a^b \frac{\rho}{2} (f(x))^2 dx}{\int_a^b \rho f(x) dx} \right)$$

Exercise 8.3.1

Suppose we have a lamina of density ρ whose shape is defined by the area between two curves. How would we find its centroid?

Exercise 8.3.2

Use the vocabulary we have discussed in this section to describe how Alexander's art in Figure 1 balances. How might you go about creating a similar piece?

8.3 Section Summary:

- We looked at the moments and center of mass of point masses in 1 and 2 dimensions.
- We extended this idea to the centroid of a continuous mass called a lamina defined by the region under a curve.